

LARA TURGUT

Physics MSc Candidate, ETH Zurich | Machine Learning & Probabilistic Modeling

Location: Paul-Feyerabend-Hof 3, HWC 43, Zurich, Switzerland

Nationality: Turkish (Swiss B Permit holder)

Work Authorization: Please see attached letter regarding work permit support.

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Web Portfolio: lturgut.github.io

GitHub: github.com/lturgut



EDUCATION

Master Studies in Physics

09/2023 – present

ETH Zurich, Zurich, Switzerland

– GPA: 5.63/6.0

– **Master's Thesis (Grade: 6.0/6.0):** *Gauge-Equivariant U-Net Architecture for Diffusion Models and Continuous Normalizing Flows*

Implemented and evaluated a gauge-equivariant U-Net architecture for generative modeling within an international research collaboration, applying diffusion models as efficient alternatives to traditional Markov Chain Monte Carlo sampling methods for complex probability distributions.

– **Master's Research Project:** *Invertible U-Net Architecture for Scalar Lattice Field Theory: A Renormalization Group Perspective*

Designed an invertible U-Net architecture within a normalizing flow framework for sampling complex probability distributions, evaluating efficiency using effective sample size.

Master Studies in Physics, Exchange Semester

09/2023 – 02/2024

EPFL, Lausanne, Switzerland

Physics BSc

09/2019 – 09/2023

ETH Zurich, Zurich, Switzerland

– **Bachelor Project:** *The Caesium Magnetometer Array of the Gradiometer for the n2EDM Experiment at PSI.*

Highschool degree (Abitur)

09/2014 – 06/2019

Istanbul High School, Istanbul, Türkiye

– Top German international high school in Türkiye with science-mathematics orientation.

– Ranked in the top 0.04% in the national high school entrance examination of Türkiye.

PRACTICAL EXPERIENCE

Teaching Assistant, Numerical Methods

02/2024 – 05/2024

ETH Zurich, Zurich, Switzerland

– Conducted exercise sessions covering numerical ODE solvers (Runge-Kutta, implicit, symplectic, and stiff methods), numerical linear algebra, and iterative methods for nonlinear equations (Newton, Broyden).

Internship, Isotope Physics Research Group

07/2023 – 08/2023

University of Vienna, Vienna, Austria

– Built a Python pipeline for data evaluation of actinide measurements using accelerator mass spectrometry.

Teaching Assistant, Physics II

02/2023 – 06/2023

ETH Zurich, Zurich, Switzerland

Teaching Assistant, Linear Algebra

09/2021 – 12/2021

ETH Zurich, Zurich, Switzerland

IT-SKILLS

Programming languages: Python (NumPy, SciPy, Pandas, Matplotlib), C++, Julia.

Machine learning: PyTorch, Scikit-learn.

Database management: PostgreSQL.

Version control: Git.

Other computer skills: LaTeX, Docker

PROJECTS

Electricity Price Forecasting for European Energy Markets

05/2026 – Present

ETH Zurich, Zurich, Switzerland

- Developing an ensemble LightGBM quantile regression pipeline for day-ahead electricity price forecasting in the DE-LU and ES electricity markets using weather, generation, and commodity features in a team.

LLM Benchmarking for Financial Reasoning with Apertus and Aisot

03/2026 – Present

ETH Zurich, Zurich, Switzerland

- Collaborating with Apertus, a Swiss LLM initiative, and Aisot, an ETH Zurich spin-off specializing in AI-driven asset management and quantitative finance.
- Benchmarking LLMs on compositional financial reasoning tasks in an interdisciplinary team.
- Designed and generated synthetic financial datasets and multi-step Q&A pairs to evaluate model reasoning capabilities.

Berezinskii–Kosterlitz–Thouless Transition in the 2D XY Model

04/2025 – 05/2025

ETH Zurich, Zurich, Switzerland

- Collaborated in an interdisciplinary team to simulate the 2D classical XY model using the Wolff cluster algorithm.

COMPETITIONS

1st Place (Team), UN OCHA Datathon Challenge

04/2026

- Challenge posed by the United Nations Office for the Coordination of Humanitarian Affairs (UN OCHA) at Switzerland's largest ML hackathon (Datathon).
- Built an explainable Bayesian model to identify under-funded humanitarian crises through a latent variable capturing systematic donor neglect, informed by needs, funding, and donor priorities.

1st Place (Team), UBS Case Competition at ETH Zurich

03/2026

- Asset management & investment banking simulation.
- Referred directly to UBS recruitment based on individual performance.

PUBLICATIONS

Diffusion Models for $SU(N)$ Gauge Theories

05/2026

J. Komijani, M. K. Marinković, L. Turgut

arxiv.org/abs/2605.06134

EXTRACURRICULAR EXPERIENCE

Board Member & Head of Marketing, Analytics Club at ETH (ACE)

09/2025 – present

ETH Zurich, Zurich, Switzerland

- Led marketing strategy and event promotion for one of Europe's largest student-run data science societies.

Marketing Committee Member, ACE

03/2025 – 09/2025

ETH Zurich, Zurich, Switzerland

Board Member & Public Relations, MindPhair

10/2023 – 10/2024

ETH Zurich, Zurich, Switzerland

- Organized outreach and communication for a career fair targeting students in mathematics, physics, and computational sciences.

Board Member & Treasurer, VMP

02/2023 – 09/2023

ETH Zurich, Zurich, Switzerland

- Managed budgeting, financial reporting, and payment operations for VMP, the Mathematics, Physics and CSE student association at ETH Zurich.

LANGUAGES

English (Fluent), German (C1 – DSD II), Turkish (Native) , Japanese (Basic).

HOBBIES

Skiing, running, gym.



WORK PERMIT ASSESSMENT

Zurich, April 29, 2026

Dear Sir or Madam,

The purpose of this letter is to confirm that the applicant, **Ms Lara Turgut**, has engaged **A-Link GmbH** to facilitate the process of obtaining a work permit in Switzerland.

Based on our professional experience in Swiss immigration and our review of Ms Turgut's CV, there is a **high chance of approval for her work permit** (for full-time fixed or permanent positions). As a holder of a Swiss university diploma, she would qualify for a simplified application process. This procedure typically takes 3 to 6 weeks and does not require proof of recruitment efforts for Swiss or EU candidates.

We offer employers the option to fully outsource the permit procedure to us. As an employer, you can benefit from:

- Regular consulting sessions tailored to the applicant and potential employer.
- Preparation and management of the Swiss work permit application.
- A-Link's direct contact with the Swiss authorities on behalf of your company.

We look forward to supporting you in launching international careers in Switzerland! For further information or inquiries, please feel free to reach out to us via our 'Get to Know Us' free call:



Sincerely yours,

Alfiya Shamisheva
CEO

ABOUT US

Launched in 2019, **A-Link GmbH** offers the easiest way to obtain Swiss work permits for highly qualified graduates, professionals, and founders. We help individuals land their dream jobs in Switzerland while supporting business growth through simplifying immigration.